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A SHACL-Based Approach for Enhancing Automated Compliance Checking with RDF Data

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Abstract: Automated Compliance Checking (ACC) has emerged as a critical tool for enforcing legal regulations across various domains. This paper contributes to ongoing research in Semantic Web technologies, particularly focusing on the execution of SHACL-SPARQL rules on RDF data. The RDF, being one of the most widely used knowledge representation (KR) formats, serves as the foundation of our approach, ensuring compatibility with existing standards and enhancing interoperability. Our research enhances the aggregate and temporal aspects of ACC by addressing the limitations of traditional ACC methodologies, which often fall short in managing the nuanced temporal and aggregate requirements essential for legal reasoning. Through a case study analysis of selected regulations with aggregate and temporal facets in LI 2204, which regulates local content and participation in Ghana's upstream petroleum industry, this paper demonstrates the effectiveness of the proposed solution in achieving these dimensions of ACC. The findings underscore the potential of Semantic Web technologies to transform ACC practices by moving towards standardized, interoperable solutions. All source codes are freely available online together with instructions to locally reproduce the simulations.

Keywords: automated compliance checking; RDF and SHACL; legal tech



Citation: Anim, J.; Robaldo, L.; Wyner, A.Z. A SHACL-Based Approach for Enhancing Automated Compliance Checking with RDF Data. *Information* **2024**, *15*, 759. <https://doi.org/10.3390/info15120759>

Academic Editor: Aneta Poniszewska-Maranda

Received: 30 September 2024
Revised: 13 November 2024
Accepted: 18 November 2024
Published: 29 November 2024



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1. Introduction

In the last decade, automated compliance checking (ACC) has received lots of attention from the scientific community [1,2], in particular from the scientific community working in Artificial Intelligence (AI) [3,4]. As a result of the gradual increase in popularity of these AI approaches, research into the automation of compliance checking has been conducted across a few domains that require compliance checking. Some of these areas include environmental regulations [5], the building and construction industry [6–9], health care design [10], and data protection [11–13].

Most of these approaches use machine learning (ML), particularly in combination with Natural Language Processing (NLP) techniques [14–17]. NLP has a prominent role in ACC because most legal knowledge is originally available in textual form (from legislation, jurisprudence, etc.); therefore, a lot of approaches propose to use legal knowledge representations in vectorized formats compatible with NLP post-analyses with respect to the in-force norms [18]. With the popularity of Large Language Models (LLMs), it is expected that, in the near future, these approaches as well as the whole research in NLP for AI and Law will be refined and extended in light of the potential offered by LLMs; preliminary investigations in that direction can be found in [19–22].

Nevertheless, since ML is based on statistical reasoning, it is difficult for it to work around *specific* and *exact* values, a requirement common in tasks such as scrutinizing compliance in compliance filing documents. This is, in particular, the case when compliance checking requires inferences on the explicitly asserted values. Thus, due to ML's statistical reasoning approach in handling represented legal knowledge, the specific or exact values which are required for inferential reasoning are often either non-existent in the representation or are not clearly captured. This makes it difficult to make inferences from the legal

knowledge represented to ascertain whether the legal rule in question has either been complied with or violated.

Another limitation of the vectorized representations typically used in ML is that their results cannot be fully explained. This is because, once they undergo training based on statistical reasoning, they often behave like a “black box”, lacking the ability to elucidate their decision-making process. Consequently, it becomes challenging, and at times even impossible, to clarify the reason why the ML system took certain decisions.

To address these limitations, *symbolic representations* have been proposed, even though they require more manual efforts, as, in contemporary approaches, symbols and rules are usually written and debugged by hand [23,24]. Symbols are suitable for dealing with specific and exact values. In addition, they correspond to human-understandable concepts, while rules represent how we reason with them; thus, the chain of logical derivations on symbols could provide intelligible explanations of AI decision making.

In the AI and Law domain, another indirect advantage that symbolic reasoning has over ML is that the former facilitates the identification of gaps within the law because of their precise and structured representation. Simply put, while ML reasons by statistical similarity and relevance, symbolic approaches adopt more fine-grained representations that work only if “all pieces of the puzzle are present and placed in the right place”. This method aids in detecting discrepancies and gaps in legal reasoning, as it explicitly represents legal concepts and their relationships as well as inherent logical inferences. ML is not adept at identifying gaps within the law since it lacks the understanding of legal concepts and it only relies on identifying recurring patterns in data. ML is effective for tasks like document classification but struggles with the nuanced nature of legal language, which is crucial in legal reasoning. In contrast, because of the explicit representation of symbolic formalization, human lawyers are able to systematically identify and address gaps within the law. An example of this advantage will be shown in Section 2.1 below: the formalization of the norms from our case study revealed to us a lacuna in the regulations, which was not so easy to identify at first glance.

To mitigate the fact that symbolic representations and rules require more manual efforts, it has been proposed to use *standardized formats*, namely W3C standards (https://www.w3.org/2001/sw/wiki/Main_Page, accessed 17 November 2024) such as RDF, SPARQL, OWL, and SHACL, which have already been successfully employed for Semantic Web in the last decades [25]. Standardized formats allow for the funneling of efforts from more people, which in turn facilitates reusability and sharing of resources. Furthermore, several NLP approaches that try to automatically create RDF knowledge graphs have also been proposed in the AI and Law domain [16,26]. With the growing popularity of LLMs, it is expected that more approaches will emerge in the near future, combining the statistical reasoning characteristic of machine learning with the logical reasoning that underpins knowledge graphs.

ACC frameworks developed with Semantic Web technologies have RDF graphs at their core [27]. Examples are [28–31], etc., which all use RDF graphs or data as the basis for their ACC frameworks, although they propose different approaches to reasoning with the RDF graphs.

This paper contributes to research on Semantic Web technology approaches for ACC, particularly in conducting aggregate and temporal reasoning on RDF graphs or data. We illustrate our approach using a case study from Ghana’s upstream petroleum industry, with a specific focus on local content and participation.

Unlike the majority of solutions proposed for automated reasoning via RDF, our representations will be based on the Shapes And Constraints Language (SHACL) (<https://www.w3.org/TR/shacl>, accessed on 17 November 2024), specifically, the new proposed extension of SHACL-SPARQL rules (<https://www.w3.org/TR/shacl-af/#SPARQLRule>, accessed on 17 November 2024).

As is well known, SHACL was not proposed for *reasoning* with RDF data, but rather for *validating* RDF data. The main W3C standard for reasoning on RDF data is instead Web

Ontology Language (OWL) (<https://www.w3.org/OWL>, accessed on 17 November 2024), and a lot of research grounded on OWL has been carried out in the last decades in AI and Law and beyond (see Section 4 below).

OWL vocabulary allows for the specification of OWL axioms that describe inferences; in particular, the main inference mechanisms in OWL are the OWL restrictions, which permit us to “restrict” the meaning of an ontological class by imposing all instances of that class to feature certain properties. In formal terms, an OWL restriction is an anonymous ontological class, defined by only describing its properties; then, by stipulating that other named ontological classes are *subclasses* of that OWL restriction, the properties defined on the latter are *inferred* for all instances of the named class via is-a inheritance [32,33].

However, it is well known that OWL is not expressive enough for representing inference rules needed in practical case studies for legal reasoning, such as the one addressed in this paper. For this reason, as will be discussed in Section 4 below, other rule-based languages, more expressive than OWL, have been used to complement or even to replace OWL, thus making the role of the W3C standard somehow vacuous.

An example of these rule-based languages is the aforementioned SHACL-SPARQL rules, although they have not yet been released as part of an official W3C recommendation, as they remain at the stage of a W3C Working Group Note. Section 4 below explains why we chose SHACL-SPARQL rules as the formal language to implement our solution.

On the other hand, for building the legal ontology for the use case, on which the SHACL-SPARQL rules for ACC are executed, we employed a mixed-methods approach to ontology engineering [34,35]. Specifically, we utilized the MeLOn (Methodology for Building Legal Ontology) methodology [11] alongside the O² validation framework from [36], which combines structural, usability, and functional dimensions for ontology assessment. Our approach adapts MeLOn to incorporate SHACL-SPARQL rules that enable automated reasoning as well as validation for compliance checking within the chosen domain.

The rest of this paper is organized as follows: Section 2 discusses the case study used in this paper to exemplify the need for formalizing compliance checking on RDF data via SHACL-SPARQL instead of OWL, while Section 3 describes the methodology we employed in developing and evaluating the legal ontology built for this case study. Section 4 then reviews the past literature on ACC for RDF data. The remainder of the paper presents the original contributions of our research. Section 5 discusses time conceptualization, or temporal management, using the Time-indexed Value in Context (TVC) ontology design pattern, as well as how we employed TVC to encode temporal dynamics in our case study. Moreover, Section 6 discusses the use of SHACL-SPARQL for inferential reasoning with RDF data and demonstrates its application in our case study. Finally, Section 7 concludes the paper and addresses future work, which involves using Large Language Models (LLMs) to semi-automatically derive them from normative texts.

2. Local Content and Participation Within Ghana’s Upstream Petroleum Sector

This section illustrates the case study used in this paper to exemplify the need of formalizing compliance checking on RDF data via SHACL rather than via OWL. Specifically, this section selects norms from the existing regulations in Ghana’s upstream petroleum industry for which the next sections will show that the expressivity offered by OWL does not suffice to represent their meaning.

The upstream petroleum industry is one sector that has contributed quite keenly to Ghana’s economy, and it is expected to contribute even further. The sector is a multibillion-dollar sector and, as a result, contributes quite significantly to Ghana’s GDP. For example, between 2013 and 2021, 4.5% of Ghana’s GDP was attributed to the oil and gas sector (<https://www.businessghana.com/site/news/business/278790/Oil-and-Gas-contributes-4-5-per-cent-to-Ghana-s-GDP-in-the-last-8-years>, accessed on 17 November 2024).

Additionally, it has been estimated that the oil and gas industry will contribute approximately GHS 15.94 billion (around 1.3 billion U.S. dollars (the dollar equivalent of

GHS 15.94 billion was calculated based on the exchange rate as at 30 November 2023)) to Ghana's gross domestic product in 2024 [37].

Given the upstream petroleum industry's economic significance, it is not surprising that it is one of the most heavily regulated sectors in Ghana. As of the last count, there are about 19 pieces of legislation governing the sector, ranging from the constitution, which is the supreme law of Ghana, to guidelines and a model petroleum agreement (<https://petrocom.gov.gh/lawsandregulations/>, accessed on 17 November 2024).

Several scholars in the domain, e.g., ref. [38], have suggested that any oil and gas producing country, in order to maximize its economic benefit from its petroleum resources, must adopt local content and participation, which is, in essence, the use and involvement of indigenous materials and personnel in the said industry.

In order to enhance local content and participation, Ghana has promulgated the Petroleum (Local Content and Local Participation) regulations 2013 (L.I 2204 (https://petrocom.gov.gh/wp-content/uploads/2022/08/13-Local-Content-and-Local-Participation-Regulations-L_I-2204.pdf, accessed on 17 November 2024)) hereinafter referred to as LI 2204 (LI 2204 has been recently amended by the Petroleum (Local Content and Local Participation) (Amendment) regulations, 2021 (L.I. 2435) https://www.petrocom.gov.gh/L&C_folder/Pet_register/laws/LI%202435.pdf, accessed on 17 November 2024). This piece of legislation regulates the local content and participation of Ghana's upstream petroleum industry.

Specifically, according to regulation 49 of LI 2204, *local content* refers to the percentage or quantum of materials, personnel, financing, goods, and services generated by indigenous companies for the upstream petroleum value chain and can be monetarily quantified (Regulation 49 of Petroleum (Local Content and Local Participation) Regulations, 2013 (L.I 2204)). Therefore, locally produced goods, services that are part of the value chain of the Ghanaian upstream petroleum industry, and anything else produced locally which is of monetary value can be classified as local content.

On the other hand, *local participation* has not been defined by the LI 2204. However, a definition may be found on the website of the Petroleum Commission; according to the said website, local participation has to do with the level of Ghanaian ownership in the upstream petroleum industry (<https://www.petrocom.gov.gh/local-content>, accessed on 17 November 2024). Therefore, the extent to which Ghanaians own companies, facilities, properties, and other assets involved in the upstream petroleum industry is what is referred to as local participation.

This is to say that the primary goal of L.I 2204 is to promote the involvement of Ghanaian citizens and the utilization of local resources within the upstream oil and gas sector. In essence, this legislation seeks to restrict foreign companies from importing their personnel and materials. Conversationally, these companies are granted permission to extract the nation's oil and gas, provided they generate employment opportunities and contribute to other economic advantages for the local population. To this end, violations of the provisions of LI 2204 and its amendment will attract sanctions (regulation 46 of the Petroleum (Local Content and Local Participation) Regulations, 2013 (L.I 2204)).

The Petroleum Commission, which is the regulatory authority for Ghana's upstream industry, is responsible for ensuring that participants in the industry abide by LI 2204 and its amendment. One of the ways specified by LI 2204 for the Petroleum Commission to check whether the participants of the industry are complying with the regulations is through the submission of compliance filing documents such as the Local Content Plan, the Local Content Performance Report, etc. The commission will then ascertain from the submitted compliance filing documents whether the participants have either complied or violated the regulations.

The compliance filing documents are currently accessible on the Petroleum Commission's website in both Excel, Word, and PDF formats. Industry participants download these files and manually complete them, submitting the filled documents to the Petroleum Commission, where they undergo manual verification. As part of his Ph.D. research, the first author of this paper is now working towards the development of a web applica-

tion to replace these Excel, Word, and PDF files. The envisioned application will enable industry participants to complete these documents online while allowing for automated or semi-automated compliance checks of the regulatory requirements.

Our framework formalizes five major regulations and three sections of the schedules in LI 2204 using the methodology discussed in Section 3. Due to the impracticality of exhaustively discussing all these provisions in this paper, we selected representative examples, which we will present in the next subsection. Most of the other regulations can be formalized in OWL or the other formal languages for legal reasoning discussed in Section 4 below. The provisions presented in the next subsection, on the contrary, cannot be formalized by any of them as only SHACL-SPARQL provides the required expressivity.

2.1. Key Provisions Motivating the Use of SHACL-SPARQL for Temporal and Aggregate Reasoning

As stated supra, participants in Ghana's oil and gas industry are required to submit compliance filing documents to the Petroleum Commission for it to check whether or not the participants are complying with the regulations. Regulation 7(2)(b) of LI 2204 states to the effect that participants (contractors, subcontractors, licensees, and other allied entities) must submit an annual Local Content Plan (LCP) to provide evidence of their compliance with LI 2204. The LCP contains several sub-plans, for the purposes of this research, such as the Employment and Training sub-plan requiring the companies to specify details of their employees. Each participant must submit its LCP every year; the Petroleum Commission establishes a window of submission that is usually opened on the first of January of the year until the last day of February. Participants are obliged to submit the LCP only within the established window of submission. Thus, a company that fails to submit its Local Content Plan commits an offense and is liable to pay to the Petroleum Commission a penalty of five percent of the value of the proceeds obtained from the petroleum activity but not exceeding the cedi equivalent of USD five million (regulation 46 (7) (B) of the Petroleum (Local Content and Local Participation) Regulations, 2013 (L.I 2204)).

The first schedule of LI 2204 provides the minimum percentages of Ghanaian employees that the participants should employ at every level of employment depending on the number of years the company has been operational. For instance, it states to the effect inter alia that (Section 3 (A), (B), and (C) of part 1 of the first schedule of Petroleum (Local Content and Local Participation) Regulations, 2013 (L.I 2204).):

- (1) a. Where the participant has been operational in Ghana for less than five years, then it must ensure that 30% of its management staff are Ghanaians.
- b. Where the participant has been operational in Ghana for five years but less than ten years, then it must ensure that 50% to 60% of its management staff are Ghanaians.
- c. Where the participant has been operational in Ghana for ten or more years, then it must ensure that 70% to 80% of its management staff are Ghanaians.

Consequently, if a company does not employ the requisite percentages of Ghanaians at any of the employment levels, that company commits an offense and is liable to pay to the Petroleum Commission an administrative penalty of two hundred thousand penalty units at best, and at worst, where the contravention persists, that company's name may be expunged from the petroleum register (regulation 46 (5) (B)(J) (K) (L) of the Petroleum (Local Content and Local Participation) Regulations, 2013 (L.I 2204)).

In order to provide evidence of compliance with respect to ((1).a–c), the annual local content plan, which contains the Employment and Training sub-plan that a participant must submit, requires the specification of all the details about the participants' employment, including employees' details, employment start dates, etc.

From this information, it is clear that OWL inferences do not suffice to check compliance of ((1).a–c), precisely for the reasons explained in the introduction: OWL lacks support for aggregate and temporal operators. The former are needed to *count* how many

employees, either Ghanaian or non-Ghanaian, the participant hired as management staff. The latter are needed to check *for how long* the participant has been operational in Ghana.

On the other hand, as already mentioned in the introduction, when trying to formalize the intended meaning of ((1).a–c), we identified a lacuna in LI 2204. The participants' status of being operational in Ghana is not a single conceptual entity that features a temporal extension, i.e., a start and an end date, as are the management staff roles. Nevertheless, LI 2204 does not specify how long the employees must be employed as management staff in relation to the window of submission for the annual LCP. Thus, a mischievous participant may decide to promote some of its Ghanaian employees on the day before the window commences and demote them the day after, just to formally comply with ((1).a–c). Even worse, the participant could "hire" new fake employees, i.e., actors, as management staff and sack them immediately after submitting the LCP. In both cases, the employees/actors that agreed to participate in the mischievous action may be bribed or rewarded with some other benefits.

Here, it is clear that even though the mischievous participant has technically complied with LI 2204, the purpose for which LI 2204 came into force, as explained above, would not be achieved. Thus, as far as the employment of Ghanaians is concerned, this blatant lacuna affords participants the avenue to manipulate LI 2204 to their own mischievous and parochial ends. To remedy this lacuna, we recommend that the parliament of Ghana consider amending LI 2204 to include, for instance, an *at least six-month* threshold of employment/promotion in relation to the start date of the window of submission for the annual LCP. In other words, it is our suggestion that the Ghanaian employees who may be counted in determining the percentages of management staff for a particular year may be those who were employed as such for *at least six months* before the start date of the window of submission for that year. In our implementation below, we will include this additional requirement to prove our point that symbolic representation can help in the identification and resolution of lacunae in legislation, although it is not currently enshrined in LI 2204. To emphasize, this is a recommendation to the parliament of Ghana to amend LI 2204 to deal with the gap identified from our formalization since it is the arm of government clothed with the authority to do so. It should be also noted that we only used the six-month threshold as an example to demonstrate our point for the purpose of this paper. The parliament of Ghana can decide on any time limit for the employment threshold if they so desire. Flowing from this, it is therefore reasonable to make the assertion that not only does symbolic representation help to efficiently achieve ACC, but it also has the capacity to aid law makers (such as the parliament of Ghana in this case) in their roles of effectively promulgating new laws and refining existing ones.

3. Adapting the MeLOn Methodology to Incorporate SHACL-SPARQL Rules

As pointed out in the previous paragraph, we built a legal ontology enriched with SHACL-SPARQL rules to represent the provisions discussed in Section 2. We used a mixed-methods approach [34–36] in developing our ontology, which currently includes the following:

- 221 classes;
- 247 properties;
- 410 individuals;
- 26 SHACL shapes;
- 48 SHACL-SPARQL rules for ACC.

While the next sections focus on the subset of ACC rules that require the expressivity of SHACL-SPARQL, this section provides an overview of the approach that we used to develop the *entire* ontology.

The starting point of our approach was the MeLOn methodology [11], a well-known methodology used for building legal ontologies. MeLOn reflects best practices in ontology engineering, as it is pivotal in achieving holistic and effective outcomes, particularly for the legal domain.

3.1. The MeLOn Methodology

The MeLOn methodology (Methodology for building Legal Ontology) (the MeLOn methodology for building legal ontologies by Prof. Monica Palmirani is described in [11]) was created by Prof. Monica Palmirani in 2015. It is an interdisciplinary approach for building legal ontologies. MeLOn addresses the unique challenges of developing ontologies in the legal domain, which is known for its complex terminology as well as intricate and interrelated concepts. By providing a structured framework, MeLOn enables experts to navigate these challenges effectively. It has been foundational in creating legal ontologies like PrOnto [11].

The MeLOn methodology is made up of ten steps as follows:

- Step 1: Provide a comprehensive description of the goal or purpose for which the legal ontology has been developed, including specific research questions or objectives and relevant use cases.
- Step 2: Define key indicators or parameters for evaluating the ontology, ensuring they align with the goals established in Step 1 and cover aspects like accuracy, completeness, consistency, and usability.
- Step 3: Investigate existing ontologies that can be integrated or linked to the ontology being developed, evaluating their applicability to save time and enhance consistency and interoperability.
- Step 4: Create a comprehensive glossary of relevant terminologies sourced from various legal documents, including definitions and links to the normative sources.
- Step 5: Develop concept tables for the ontology, categorizing elements into classes, object properties, datatype properties, and OWL restrictions.
- Step 6: Utilize tools like Protégé (<https://protege.stanford.edu>, accessed 17 November 2024) or yED (<https://www.yworks.com/products/yed>, accessed 17 November 2024) for the development of the ontology.
- Step 7: Test the ontology continuously during development.
- Step 8: Refine the ontology based on the testing phase, addressing identified errors and expanding its scope to include all necessary elements.
- Step 9: Evaluate the ontology against the parameters established in Step 2 to ensure it meets the predefined criteria and standards.
- Step 10: Publish the ontology, making it publicly accessible, integrating it with other existing ontologies, and documenting it comprehensively.

The MeLOn methodology, like most other methodologies, was primarily designed for OWL. However, since our approach significantly incorporates SHACL-SPARQL rules in the ontology engineering process, not all ten steps were followed to the letter. For instance, we did not use Protégé because it does not support SHACL-SPARQL. Rather, we developed our own tools to execute the rules; some code from these tools is available in the GitHub repository associated with this paper.

The forthcoming PhD dissertation by the first author of this paper, currently being written, will cover all ten steps in detail. Some of these details are included here as well (e.g., Step 1 is partially covered in Section 2 above); however, most of these details are far beyond the scope of the present paper and are therefore omitted.

The next subsection will just provide some additional information about Step 2, specifically detailing the approaches that we used to build and evaluate our ontology.

3.2. MeLOn, Step 2

The evaluation of ontologies has been a fundamental part of ontology development since the early 1990s, as discussed in [39]. Despite ongoing advancements, the literature on ontology evaluation remains fragmented and lacks a unified approach. Various evaluation methods have emerged, often focusing on specific issues or multiple quality criteria simultaneously, which has left the literature unclear on core challenges and best practices. Most

evaluation efforts prioritize functionality and performance over examining the structural and compositional aspects of ontologies [36].

Several evaluation methods, such as OntoMetric [40], OntoClean [41], and O² [36], are used for ontology assessment. Among these, O² was chosen as the primary evaluation framework for this work, as it organizes evaluation indicators into three core dimensions: structural, usability, and functional. This framework guided the evaluation process for the legal ontology, establishing a systematic foundation and clarifying the principles underlying the assessment criteria, thereby promoting transparency in the methodology.

3.2.1. Structural Dimension

The structural dimension of an ontology focuses on its graph representation, capturing both topological and logical aspects independently of specific metrics. Topological properties refer to the structure of the ontology's graph, such as depth, breadth, connectivity, and density, which graph-based metrics quantify to reveal its complexity, modularity, and navigability. Logical properties, in contrast, deal with the ontology's formal semantics, ensuring consistency, inference potential, and compliance with established standards. While metrics offer quantitative insights, structural dimensions are intrinsic to the ontology, shaping its design, organization, and usability regardless of the metrics used [36].

Hierarchies of classes, organized through a superclass–subclass (*is-a*) relationship, form the backbone of ontologies' structure [42–45]. To construct these hierarchies, two main methods are typically employed: the top-down approach, which starts from general concepts and refines them into specific subclasses, ensuring logical consistency and comprehensive domain understanding [46]; and the bottom-up approach, which begins with specific concepts and generalizes them into broader classes, particularly useful for well-defined, instance-rich domains [47]. A combined approach was used to develop our ontology's class hierarchy. Beginning with top-down phases to establish general concepts as root nodes, then progressing to bottom-up phases to identify specific instances and generalize them into subclasses, this hybrid strategy iteratively alternated between both approaches. This integration created a balanced, comprehensive class hierarchy that captures domain complexity accurately and coherently [48]. Furthermore, the ontology adheres to the design patterns of the imported ontologies, among which is TVC, which we used to represent ontological entities with a temporal dimension. Given its relevance to the research presented here, TVC will be explained in detail below in Section 5. Additionally, we developed tools to ensure the ontology remained logically consistent, as well as semantically and syntactically accurate, throughout its development.

3.2.2. Usability Dimension

The usability dimensions of an evaluation parameter are heavily influenced by the level of annotation and documentation provided. A well-annotated ontology makes it easier for users to understand the properties, classes, and relationships within the ontology, as well as how they relate to real-world concepts. Without proper annotation, users may struggle to recognize the intended meaning and purpose of the ontology's components [36]. In light of this, we annotated almost all the properties, classes, and individuals in our ontology through the RDF annotation property `rdfs:comment`. These comments defined and, in some cases, explained the said classes, properties, and individuals, thus providing proper documentation to understand the legal ontology. This paper and the forthcoming PhD dissertation by the first author provide further in-depth explanations of the RDF resources in the ontology and the SHACL-SPARQL rules for ACC.

3.2.3. Functional Dimension

The functional dimension of an ontology pertains to its purpose and the roles of its components, influenced by the context of its creation and use. In general, ontologies focus on knowledge sharing and emphasize terminological agreement; however, ontologies

designed for automation and decision support, such as the one we developed, prioritize reasoning, rule encoding, and integration capabilities [36].

In light of this, we employed a task-based approach to assess the legal ontology's functional dimension. The task-based approach is application-oriented, measuring how well an ontology enhances the performance of a designated task [34]. Unlike structural evaluations that examine conceptual hierarchy and adherence to best practices, this approach emphasizes practical, outcome-driven assessments of task support, adaptability, and tool compatibility. Task-based evaluations also analyze task speed and identify inconsistencies by observing performance [34]. For instance, [49] applied this approach to evaluate an ontology's utility by measuring user effort through a complexity-based cost model.

A common approach to task-based evaluation, which we employed in our work, involves listing, in advance, the Competency Questions (CQs) that the system is expected to answer correctly [48]. The next subsection lists the CQs that we postulated a priori with respect to the norms from LI 2204 considered in this paper.

3.2.4. Competence Questions (CQs)

CQs are essential for defining the requirements of ontologies and are crucial to ontology development [48,50–52]. They represent inquiries that the ontology should be able to answer, determining what is relevant for the ontology. CQs help outline its scope and provide a method for evaluation, supporting both the specification and assessment of an ontology [53]. In the specification phase, CQs assist in modeling the domain by highlighting relevant concepts and their relationships. During evaluation, they help identify flaws in domain modeling, contributing to the ontology's overall quality [46]. The aim of CQs is to ensure that ontologies accurately reflect the real world [53].

Below are three examples of the CQs we defined for the ontology in Step 2, i.e., *before* designing and implementing the ontology. CQ3 relates to one of the provisions in (1).

- CQ1: Did the contractor, subcontractor, licensee, or allied entity submit its annual local content plan? If it did, was the submission within the window of submission stipulated by the Petroleum Commission? (Regulation 7 (2) (B) Petroleum (Local Content and Local Participation) Regulations, 2013 (L.I 2204).)
- CQ2: Did the contractor, subcontractor, licensee, or allied entity retain only the services of a Ghanaian legal practitioner or a firm of Ghanaian legal practitioners for its petroleum activity that requires legal services in the country? (Regulation 29 Petroleum (Local Content and Local Participation) Regulations, 2013 (L.I 2204).)
- CQ3: Was at least 50% to 60% of the management staff of the contractor, subcontractor, licensee, or allied entity composed of Ghanaians during the period between the first five years and before the ten-year mark of its operation in the country? (Section 3 (A) of part 1 of the first schedule of the Petroleum (Local Content and Local Participation) Regulations, 2013 (L.I 2204).)

4. Related Approaches

As pointed out above in the introduction, this paper argues that W3C standards have to be used for ACC based on symbolic reasoning, given their widespread adoption and, consequently, their high potential for interoperability and reusability.

This section reviews the literature on ACC with RDF data. As mentioned in the introduction and exemplified in the previous section, since temporal reasoning is critical to our case study and, indeed, to the majority of real-world compliance checking tasks, the next section will introduce the Time-indexed Value in Context (TVC) (<https://sparontologies.github.io/tvc/current/tvc.html>, accessed 17 November 2024), a well-known *ontology design pattern* used to represent conceptual entities with temporal extensions.

This paper is not the first to address compliance checking on RDF data. Previous approaches have used RDF/OWL to model the TBox and states of affairs. However, as already observed earlier, OWL is well known for its limitations in expressiveness.

OWL's limited expressiveness for legal reasoning has primarily been attributed to its lack of mechanisms for handling defeasibility. For instance, the approach in [54] encodes legal rules within OWL2 decidable profiles to maintain control over computational complexity. In this approach, norms are represented as property restrictions that refer to subsets of individuals who comply with the norms, with compliance checking enforced through OWL2 subsumption. However, the authors themselves acknowledge (see [54], Section 3.3) that their approach is limited to GDPR policy validation and it does not truly involve legal reasoning, which is inherently defeasible. Ref. [54] points to the framework presented in [55] as an example of full legal reasoning for GDPR compliance. Ref. [55] employs the SPINdle reasoner [56] to perform the necessary (defeasible) inferences. However, as recently documented in [2], the SPINdle reasoner has significant limitations in usability and scalability, which is instead the main focus of [54].

Ref. [55] is not the only approach that addresses OWL's lack of expressiveness. Several others have been proposed, all utilizing separate knowledge bases with rules encoded in alternative formats to represent norms that OWL cannot express adequately. Some of these alternative formats are: SWRL (<https://www.w3.org/Submission/SWRL>, accessed 17 November 2024) [57], LKIF-rules [58], RuleML [59], and LegalRuleML (<https://www.oasis-open.org/committees/legalruleml>, accessed 17 November 2024), which is the one used in [55].

However, in our view SHACL-SPARQL is best suited for compliance checking in comparison to other rule- or ontology-based approaches such as SWRL or LegalRuleML. SHACL-SPARQL subsumes SWRL, which itself was subsumed under the OASIS standard of RuleML [59]; it is the successor to the SPARQL inferencing notation (SPIN) (<https://spinrdf.org>, accessed 17 November 2024). LegalRuleML is a specialization and augmentation of RuleML; it aims at representing and interchanging legal norms. LegalRuleML is not itself designed for compliance checking, as it does not prescribe any formal semantics but requires transformations from LegalRuleML into a machine-readable reasoning language, which is then provided to a defeasible reasoning engine, e.g., the SPINdle reasoner [55,56].

In light of this, Ref. [60] made a preliminary proposal to extend the LegalRuleML meta model [61] and to represent normative rules via SPARQL queries. Ref. [60] is, to our knowledge, the first proposal that models ACC by employing W3C standards only. The solution presented in this paper is therefore rather close to [60]'s; however, while [60] only uses SPARQL, our formalization will use SHACL-SPARQL.

Similarly to [54,62,63], we distinguish compliant and non-compliant individuals by introducing, respectively, special OWL subclasses and special SHACL shapes. Contrary to [54], however, Refs. [62,63] can model defeasible inferences.

For instance, the OWL ontology in [62] includes two classes *Supplier* and *Vehicle*. The individuals in *Supplier* are obliged to communicate their contractual conditions to their consumers (R1), while vehicles cannot drive over 90 km/h (R2). To implement R1 and R2, *Supplier* and *Vehicle*, respectively, include a boolean data-type property *hasCommunicatedConditions* and a float data-type property *hasDrivingSpeed*. Then, two subclasses *SupplierR1compliant* and *VehicleR2compliant* are defined, the former including individuals in *Supplier* for which *hasCommunicatedConditions* is true, the latter including individuals in *Vehicle* for which *hasDrivingSpeed* is lower than 90. Compliance checking is then enforced by simply applying OWL2 subsumption. In other words, OWL "is-a" inferences will populate *SupplierR1compliant* and *VehicleR2compliant* with only the individuals that comply with the two norms.

In this setting, exceptions may be added by defining complement subclasses via the OWL2 tags at disposal, e.g., *owl:disjointWith*. These subclasses will define the subsets of individuals that *violate* the norms; thus, by imposing the set of the individuals that comply with a norm as *owl:disjointWith* on the one that violates it, the correct inferences are achieved.

The solution in [63] is very close to the one of [62], the crucial difference being that [63] uses SHACL shapes in place of OWL2 subclasses/restrictions to validate the values of the relevant attributes, e.g., `hasCommunicatedConditions` and `hasDrivingSpeed`, in the example above. In addition, SHACL Triple rules are used to compute the values of these attributes. These rules collect partial data from the RDF triples, and so they facilitate the representation of the norms by decoupling it in multiple sequential steps. Once the SHACL Triple rules have computed the values of the attributes, the SHACL shapes are executed in order to validate these values. Individuals with invalid attribute values are labeled as non-compliant. Finally, defeasibility is modeled via negation-as-failure: whenever exceptions hold, corresponding SHACL Triple rules defeat other ones, so that the inferences associated with the latter are blocked.

Among the approaches mentioned above, only those that support SPARQL are expressive enough to represent the norms in ((1).a–c). SPARQL is the single mentioned formal language that supports aggregate and temporal operators. However, none of the approaches seen above discuss aggregations and temporal management, despite their crucial role in compliance checking and accountability, while this is what this paper will specifically address, as mentioned in the introduction.

SHACL-SPARQL rules have been proposed to extend SHACL with inference capabilities, aiming towards the definition of a single unified W3C standard for *both* validation and inference [9,64,65].

As the name suggests, SHACL-SPARQL extends SHACL with some functionalities of SPARQL. SHACL-SPARQL rules, in particular, are SPARQL queries in the form `CONSTRUCT-WHERE`, which makes it possible to add additional triples *inferred* from the ones already asserted in the knowledge graph.

Therefore, SHACL-SPARQL rules behave as a standard *production rules system*: for all RDF triples matching the `WHERE` clause, the `CONSTRUCT` clause builds further triples and adds them to the knowledge graph. Note that the rules are monotone, as only the `CONSTRUCT` clause is allowed; in other words, triples can never be deleted or updated.

Regarding scalability with this type of rule, ref. [66] recently demonstrated that efficiency can be achieved by using the Datalog and Answer Set Programming reasoner DLV2 [67], which supports embedding SPARQL queries. Based on the results in [66], the solution proposed in this paper appears then to be optimal both in terms of usability, as SPARQL is widely adopted both due to the rapid expansion of the Semantic Web and to its simple and intuitive syntax, and in terms of scalability, since the rules can be integrated into the DLV2 reasoner, which is highly efficient.

Given the highlighted SHACL-SPARQL's peculiarities, this paper extends the research directions initiated in [2] by demonstrating how the regulations discussed in the previous section can be represented using SHACL-SPARQL. These norms involve temporal reasoning and aggregate functions, which are extremely challenging, if not impossible, to represent in the other rule-based formats mentioned earlier. To the best of our knowledge, no prior attempts have been made to model these norms in those formats.

However, before presenting our implementation in SHACL-SPARQL, it is necessary to understand how it is possible to represent time-stamped knowledge in RDF, i.e., knowledge that features a start and an end date, which are relevant to temporal reasoning. In this paper, as mentioned above, we follow the Time-indexed Value in Context (TVC) ontology design pattern, which will be discussed in the next section.

5. Time-Indexed Value in Context (TVC)

In our case study, time plays an important role in determining whether a participant in the upstream petroleum industry is compliant with LI 2204. Hence, for ACC to be achieved in the area of local content and participation of Ghana's upstream petroleum industry, the temporal aspects must be represented. For instance, an employee of a company can play different roles at different times within his or her employment period, as we discussed in Section 2.1 above.

SHACL-SPARQL does not suffice to deal with the complexities of time management. SHACL-SPARQL is a language for querying and reasoning with RDF data, it is not concerned with how to represent knowledge in RDF, i.e., how to properly represent features and values of conceptual entities that vary over time.

The Time-indexed Value in Context (TVC) ontology design pattern has instead been proposed to this end. We chose to use it to represent our case study.

TVC, described as an extension of the Time-indexed Situation pattern (<http://www.ontologydesignpatterns.org/cp/owl/timeindexedsituation.owl>, accessed 17 November 2024), is an OWL ontology providing a comprehensive and flexible framework for describing instances where entities have values (e.g., roles, statuses) within contexts and during specific time frames [68]. Its classes and properties allow specifying values that conceptual entities have in certain contexts that vary over time, enabling representation of time periods during which entities had particular values, as well as the contexts within which entities have those values, which is significant since values may depend on the considered context, e.g., an employee being a foreign employee in a subsidiary but indigenous in the parent company.

The main classes and properties of the TVC ontology design pattern are shown in Figure 1. The central class is `ValueInTime` class, which defines a specific type of temporally organized scenario that serves as a focal point connecting an entity with a specific value, encompassing the value itself, and considering the temporal and contextual boundaries that influence the relationship between the entity and the value (the TVC ontology design pattern figure from <https://sparontologies.github.io/tvc/current/tvc.html>, accessed 17 November 2024).

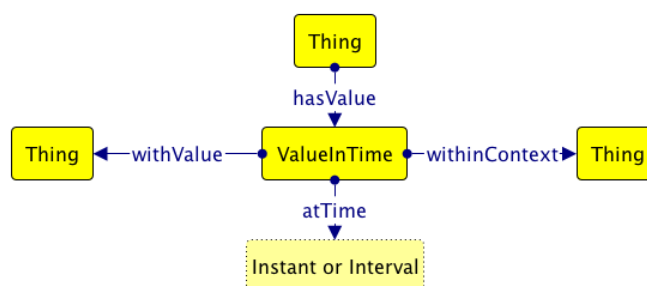


Figure 1. TVC ontology design pattern.

TVC includes four main object properties. One of these, `hasValue`, specifies in its range the class `ValueInTime`: this property is used to associate an individual of the state of affairs to a particular value in time and context. We will use this property to, for instance, associate the employees with their role; the latter, as explained above, is a value in time (i.e., it has a start and end date), in the context of a particular company.

The other three object properties relate the value in time and context with the features describing it; therefore, the class `ValueInTime` is specified in their domain rather than in their range. These object properties are as follows:

- `atTime`, which defines the individual `Instant` or `Interval` in which the entity has a value. The classes `Instant` or `Interval` are part of the Time Ontology (<https://www.w3.org/TR/owl-time>, accessed 17 November 2024), which is imported in TVC.
- `withValue`, which defines the particular value, e.g., the specific role that an employee has within a company, held by the entity.
- `withinContext`, which defines the relevant context in which an entity holds the value, e.g., the company in which the employee is employed.

TVC makes it possible to represent all the employment details of an employee within different periods and contexts. Through TVC, we can represent the employment period of a Ghanaian employee, the various roles played by that employee within that period, and at what times these roles were played within the employment period.

Besides TVC, our formalization below will use another well-known ontology design pattern: Time Interval (<http://www.ontologydesignpatterns.org/cp/owl/timeinterval.owl>, accessed 17 November 2024). Time Interval is a simple ontology design pattern that only includes the class `TimeInterval` and the two datatype properties `hasIntervalStartDate` and `hasIntervalEndDate` that relate instances of `TimeInterval` with a date.

Of course, it must be checked that the start date precedes the end date. However, Time Interval does not implement this check because it is based on OWL, which is unable to compare dates. We therefore added the following SHACL shape to our ontology, for invalidating instances of `TimeInterval` in which the end date precedes the starting one:

```
(2) :CheckDatesTimeInterval rdf:type sh:NodeShape;
    sh:sparql[sh:prefixes ...;
        sh:message "Interval {${this}} is invalid: the starting date {?sd}
                    comes after the end date {?ed}";
        sh:select """SELECT ${this} ?sd ?ed
                    WHERE{${this} ti:hasIntervalStartDate ?sd.
                        ${this} ti:hasIntervalEndDate ?ed.
                        FILTER(?sd>?ed).}""";
    sh:targetClass ti:TimeInterval.
```

Time Interval is much simpler to use than the Time Ontology; therefore, our formalization below will set `TimeInterval` as a subclass of the class `Interval` of the Time Ontology in order to use the individuals belonging to the former as objects of TVC's `atTime` property. Furthermore, since the Time Interval ontology design pattern does not specify the ranges of its two datatype properties (see <http://www.ontologydesignpatterns.org/cp/owl/timeinterval.owl>, accessed 17 November 2024: "Ranges of datatype properties are not indicated, because otherwise specializing them with different xsd datatypes would lead to inconsistencies"), we chose in our implementation to set these ranges to `xsd:dateTime`. Further SHACL shapes have been added to check, but this paper omits them; the interested reader might find them on the GitHub repository.

Representing Employees' Roles in the Case Study

As stated earlier, TVC and Time Interval made it possible for us to represent the employment details within different periods and contexts. This section shows how we represented employment roles played by employees within their employment periods.

As indicated supra, every employee has an employment period within which he/she plays his/her employment role. Thus, the representation of the employment role of an employee requires the representation of the employment period of that employee, which will act as context for the employment role. We used the following class to represent the employees' employment periods:

- `EmploymentPeriod`, which is a subclass of `tvc:ValueInTime`. In our knowledge graph, the value of an employment period of an employee is the upstream oil company within which the employee is employed.

The following properties relate the employees with their employment periods and specify the values at times of the latter. Note that, on the other hand, we do not *explicitly* represent the context of the employment periods as this is already *implicitly* assumed as the upstream petroleum industry of Ghana.

- `has-employment`, which is a sub-property of `tvc:hasValue`. This property defines the relationship between an employee and an employment period value in time instance. Thus, our ontology sets its domain and range as the classes `Employee` and `EmploymentPeriod`, respectively.
- `with-company`, which is a sub-property of `tvc:withValue`. It relates the employment period value in time instance to the company in which the employee worked/works within his or her employment period.

- `at-time-of-employment`, which is a sub-property of `tvac:atTime`. This property defines the time period of the employee's employment. Thus, it relates the employment period value in time instance to its respective `ti:TimeInterval` instance.

An example of an employee's employment period, taken from the knowledge graph available on the GitHub repository, is the following:

```
(3) :sheldonCooper :has-employment :sheldonENIEmployment.
    :sheldonENIEmployment rdf:type :EmploymentPeriod;
    :with-company :egepl;
    :at-time-of-employment :sheldonENIEmploymentPeriod.
    :sheldonENIEmploymentPeriod rdf:type ti:TimeInterval;
    ti:hasIntervalStartDate "2016-07-07T00:00:00Z"^^xsd:dateTime.
```

The individual `egepl` refers to “ENI Ghana Exploration And Production Limited”, which is also an International Oil Company currently operating in Ghana as a contractor.

Rule (3) encodes that Sheldon Cooper (Of course, all employees' names used in our examples are invented.) is an employee of `egepl`. He was employed on the 7 July 2016. Conversely, the end date of the employment period is not specified; in our knowledge graph, it is assumed that when end dates of employment periods (or employment role periods) are not specified, the employee, e.g., Sheldon Cooper in this example, is still employed by the companies, e.g., `egepl` in this example.

Having looked at Sheldon's Cooper's employment period, we will now turn our attention to his corresponding employment role within that period. To represent the employment role periods of the employees, we introduced the following class:

- `EmploymentRolePeriod`, which is a subclass of `tvac:ValueInTime`. It encodes the roles played by the employees within the context of their employment periods.

The following properties have been defined to relate the employees with their employment role periods and to specify the value, time, and context of the latter:

- `has-employment-role`, which is a sub-property of `tvac:hasValue`. It associates an employee with an employment role within a particular context and timeframe.
- `with-role`, which is a sub-property of `tvac:withValue`. It associates an employment role value in time to the particular role (management staff, technical core staff, etc.) played by an employee.
- `within-employment-context`, which is a sub-property of `tvac:withinContext`. It specifies the employment period in which an employee plays his or her employment role. In other words, this property stipulates the employment period as the context within which an employee's role is played.
- `at-employment-role-time`, which is a sub-property of `tvac:atTime`. This property defines the timeframe within which an employee plays his/her employment role.

The representation of Sheldon Cooper's employment role is the following:

```
(4) :sheldonCooper :has-employment-role :sheldonENIEmploymentRole.
    :sheldonENIEmploymentRole rdf:type :EmploymentRolePeriod;
    :with-role wd:Q1210069;
    :within-employment-context :sheldonENIEmployment;
    :at-employment-role-time :sheldonENIEmploymentRolePeriod.
    :sheldonENIEmploymentRolePeriod rdf:type ti:TimeInterval;
    ti:hasIntervalStartDate "2016-07-07T00:00:00Z"^^xsd:dateTime.
```

In (4), `wd` is the prefix of Wikidata; thus, `wd:Q1210069` refers to the Wikidata concept “general manager” (<https://www.wikidata.org/wiki/Q1210069>, accessed 17 November

2024). Rule (4) encodes that Sheldon Cooper has a general managerial role within the company egepl from the 7 July 2016, i.e., from the same date in which he was hired. This means that Sheldon Cooper’s employment period began with his role as the general manager of the company, and since there is equally no end date for his employment role, he still plays that role.

On the other hand, other employees of the upstream petroleum companies could be employed to play a certain role and later promoted or demoted to play another one. In our knowledge graph, these employees will be associated with multiple instances of the class EmploymentRolePeriod, which will all have to fall within the timeframe of the EmploymentPeriod instance specified in the within-employment-context property. In order to enforce this check, we wrote the following SHACL shape for our ontology:

```
(5) :ERPWithinEP rdf:type sh:NodeShape;
    sh:sparql[sh:prefixes ...; sh:message "{$this} is invalid: the
        employment role period must be within the employment period";
    sh:select """SELECT $this ?ert ?erts ?toes
        WHERE {$this :at-employment-role-time ?ert.
            $this :within-employment-context ?c.
            ?ert ti:hasIntervalStartDate ?erts.
            ?c :at-time-of-employment ?toe.
            ?toe ti:hasIntervalStartDate ?toes.
            FILTER(?erts<?toes).}""";
    sh:sparql[sh:prefixes ...; sh:message "{$this} is invalid: the
        employment role period must be within the employment period";
    sh:select """SELECT $this ?ert ?erte ?toee
        WHERE {$this :at-employment-role-time ?ert.
            $this :within-employment-context ?c.
            ?ert ti:hasIntervalEndDate ?erte.
            ?c :at-time-of-employment ?toe.
            ?toe ti:hasIntervalEndDate ?toee.
            FILTER(?erte>?toee).}""";
    sh:targetObjectsOf :has-employment-role.
```

The SHACL shape above ensures that an employment role period is fully contained within the corresponding employment period. This is achieved through two SPARQL constraints. The first checks that the start date of the employment role period is on or after the start date of the employment period, while the second verifies that the end date of the employment role period is on or before the end date of the employment period. If either condition is violated, an error message indicates the employment role period is outside the defined employment period.

Besides Sheldon Cooper, the knowledge graph includes seven other individuals who are either Ghanaian citizens or foreigners with various nationalities split evenly between the employment of Tullow Ghana limited and ENI Ghana Exploration And Production Limited, hereinafter referred to as Tullow and ENI, respectively. All individuals are listed below in Tables 1 and 2, together with their nationalities, employment periods, the company they work for, their roles within the company, and the period of these roles. In Tables 1 and 2, “EP” is short for “Employment Period”, “ER” for “Employment Role”, and “ERP” for “Employment Role Period”. Note that “Ukuuk Mina” played two different employment roles within his employment period (the RDF encoding of the data in the two tables is available on the GitHub repository).

Table 1. Individuals in the knowledge graph are associated with their employer, nationality, and the start and end dates of their employment periods. The label “n/a” indicates when data are unspecified; thus, if no end date is provided for any employee, it implies that they are still employed by the company, as reflected in the second column.

Name	Company	Nationality	EP Start Date	EP End Date
Ukuuk Mina	ENI	Ghanaian	07/07/2016	n/a
Paolo Rossi	ENI	Italian	01/12/2017	n/a
John Smith	Tullow	American	31/12/2010	n/a
Leonardo Davinci	ENI	Italian	07/07/2016	n/a
Sheldon Cooper	ENI	American	07/07/2016	n/a
Kwame Anim	Tullow	Ghanaian	31/12/2022	n/a
Efua Fiankobe	Tullow	Ghanaian	31/12/2022	n/a
Yaa Adubea	Tullow	Ghanaian	31/12/2022	n/a

Table 2. Individuals of the knowledge graph together with their role and employment role period’s start and end dates. As in Table 1, “n/a” marks when the data are not specified; furthermore, all values in the ER column are individuals of the class ManagementStaff.

Name	ER	ERP Start Dates	ERP End Dates
Ukuuk Mina	CFO; COO	07/07/2016; 07/07/2017	06/06/2017; n/a
Paolo Rossi	CEO	01/12/2017	n/a
John Smith	CMO	31/12/2010	n/a
Leonardo Davinci	CMO	07/07/2016	n/a
Sheldon Cooper	General Manager	07/07/2016	n/a
Kwame Anim	CEO	31/12/2022	n/a
Efua Fiankobe	COO	31/12/2022	n/a
Yaa Adubea	COO	31/12/2022	n/a

6. Checking Compliance via SHACL-SPARQL Rules

Section 2.1 above describes the regulations as stipulated in LI 2204, which we chose for the use case of this paper. In order to implement ACC within the local content and participation domain of the upstream petroleum sector of Ghana, we encoded, in line with the recent literature (cf. Section 4 above), the chosen regulations into SHACL-SPARQL rules because of their expressivity and logical reasoning capabilities.

In particular, Section 4 above explained that the solution proposed in this paper is rather close to the solution proposed in [60]: both solutions are based on SPARQL, which make them capable of handling aggregate and temporal information. On the other hand, SPARQL alone is unable to prioritize the rules while SHACL-SPARQL vocabulary includes the property `sh:order`, which allows us to specify the execution order of the rules, thus allowing for (some) control flow within the framework. Specifically, `sh:order` is a datatype property which associates rules with integers, and rules are then executed from the lowest to the highest value specified in `sh:order`; in cases where two rules are associated with the same value, their order of execution is randomly selected by the automated reasoner. In essence, `sh` enables the implementation of procedural programs that use SPARQL rules, of which our proposed approach is no exception.

To achieve ACC in complex use cases, e.g., our own, certain rules must be executed before others; thus, SPARQL alone does not provide the required expressivity.

For instance, in Section 2.1, it was stated in effect that compliance checking can only be conducted if the Petroleum Commission has received the main compliance filing document,

which is the Local Content Plan (LCP). Thus, without the LCP, the commission cannot conduct the necessary compliance checks. As a result, companies are obligated to submit their LCP within a particular time window. Hence, in implementing an ACC framework for this domain, the first rules to be executed must check whether the LCP plan has been submitted; if so, it has to subsequently check whether it has been submitted within the allowed time window. If a company has already violated these requirements, no further compliance checks can be conducted.

It would be rather complex, if not impossible, to implement these rules via a single SPARQL rule as proposed in [60]. Thanks to `sh:order`, on the other hand, it is easily possible to *decouple* complex compliance checks into a sequence of simpler SPARQL rules.

Furthermore, in the proposed framework, rule prioritization can be utilized to implement conflict resolution through rule overriding, which is essential in legal reasoning. It is, in fact, quite common to find specific rules in legislation that override general rules in particular contexts. However, conflict resolution falls outside the scope of this paper, which focuses on aggregative and temporal reasoning. We refer interested readers to [2,66], which discuss SHACL-SPARQL rules for managing rule overriding through priorities, fully compatible with the rules presented in this paper.

6.1. Checking Constraints Including Temporal Constraints

This subsection presents the formalization of the selected norms from our use case. However, before we proceed, it must be clarified that the goal of the developed production rule system is to identify companies that did *not* comply with the regulations, so they can be penalized in accordance with regulation 46 of LI 2204 (regulation 46 of the Petroleum (Local Content and Local Participation) Regulations, 2013 (L.I 2204)). Therefore, we designed the rules to identify companies that violate specific compliance checks, flagging them as “non-compliant” during the execution process. This is carried out by adding the companies within a special class `NonCompliantCompany` while associating them with an explanation in plain text through the property `rdfs:comment` (see examples below).

The rules are executed sequentially. The sequence ensures that a company is only subjected to a rule if it has not already been flagged as “non-compliant” by any of the preceding rules, thereby optimizing the overall process.

The first rule, having `sh:order` equal to 0, searches for companies that did *not* submit their LCPs at all. This rule goes through the provided ABox of the ontology to identify all companies which have not submitted an LCP. It then flags all such companies as “non-compliant”.

As specified *supra*, if a company is flagged as “non-compliant”, no further checks can be conducted on it.

On the other hand, companies that pass the first check will be subsequently checked by the second rule, having `sh:order` equal to 1. Given this value on the `sh:order` property, this rule will only be executed after the previous one. In order to understand whether the companies passed the first check, this rule will check whether they were *not* flagged as “non-compliant”.

For all such companies, the second rule checks whether they submitted their LCPs *outside* the time frame within which the LCPs were meant to be submitted. The rule goes through the provided ABox of the ontology to identify the actual submission dates of the LCPs by these companies in order to check whether they fall within or outside the stipulated window of submission. As we have seen above in the SHACL shapes (2) and (5), SPARQL, contrary to OWL, allows for the comparison of `xsd:dateTime` values. If the rule identifies any submission date that falls outside the window of submission, it again flags the company as “non-compliant”.

It should now be clear to the reader how we decided to implement control flow on the different rules. They are executed in a sequence, by setting incremental values on the `sh:order` property. Rules search for companies that *violate* the implemented checks, by flagging them as “non-compliant” as long as the rules are executed. Rules in the

sequence are executed on a company only if that company did not violate any check from the previous rules.

Finally, in order to distinguish the several reasons of non-compliance, each rule will also associate the non-compliant company with an `rdfs:comment` explaining in plain text why the company has been flagged as non-compliant, thus providing a basic degree of *explanation* for the rule's outcome. These comments are of course intended to be read a posteriori.

Let us now show how these rules are specifically encoded in SHACL-SPARQL. However, before that, the classes and properties used therein need to be introduced.

The following RDF properties were used in the WHERE clause of the first rule. They are used to encode the conditions that must be met for the triples in the CONSTRUCT clause to be added to the inferred ABox.

- `has-submitted` relates a company to its submitted LCP.
- `plays-upstream-company-role`, which is a sub-property of `tvcl:hasValue`. This property relates a company with an individual of `UpstreamCompanyRolePeriod`, which is a value in time and context specifying the role that the company plays within a specific timeframe. For instance, for the individual `tgl`, which denotes the other company we consider in our use case, i.e., Tullow, it holds:

```
:tgl :plays-upstream-company-role :tullowContractor.
:tullowContractor rdf:type :UpstreamCompanyRolePeriod;
:with-company-role :contractor;
:at-contractor-role-time :tullowContractorPeriod.
:tullowContractorPeriod rdf:type ti:TimeInterval;
ti:hasIntervalStartDate "2006-03-10T00:00:00Z"^^xsd:dateTime;
ti:hasIntervalEndDate "2031-03-10T00:00:00Z"^^xsd:dateTime.
```

in which `contractor` is formalized as:

```
:contractor rdf:type :UpstreamCompanyRole;
rdfs:comment "a set of oil companies that have entered into a
petroleum agreement with Ghana to undertake petroleum exploration
and production activities under Act 919".
```

Note, on the other hand, that `tullowContractor` does not specify the context; this is implicitly set to the upstream petroleum industry of Ghana.

With the introduced properties, the first rule is implemented as follows:

```
(6) sh:rule[rdf:type sh:SPARQLRule; sh:order 0; sh:prefixes ...;
sh:construct """
CONSTRUCT{NonCompliantCompany rdf:type owl:class.
?x rdf:type :NonCompliantCompany;
rdfs:comment "This oil company or service provider
is not compliant because it did not submit its local
content plan to the Petroleum Commission."}
WHERE{?x :plays-upstream-company-role ?r.
NOT EXISTS{?x :has-submitted ?z}}"""]
```

The reader can verify, by downloading and locally re-executing the code on GitHub, that both individuals `egepl` and `tgl`, who, respectively, denote the companies ENI and Tullow, comply with the check in (6) because they both submitted their LCP.

To implement the second rule, we further introduced the class and the property below:

- `SubmissionWindowLCP`, which is a subclass of `TimeInterval`. It defines the time window within which all annual LCPs must be submitted for a particular year.
- `at-submission-time`, which specifies the `xsd:dateTime` values of the LCPs submitted by the companies.

With these two further properties, the second rule is implemented as in (7). Note that the `sh:order` value of the rule (7) is superior to the one of (6), meaning that it will be executed after the latter. Note also that the first check implemented in (7) is whether the company already belongs to the class `NonCompliantCompany`; only if the company does not belong to it, i.e., only if the `NOT EXISTS` clause is satisfied, are the other checks carried out.

```
(7) sh:rule[rdf:type sh:SPARQLRule; sh:order 1; sh:prefixes ...;
sh:construct """
CONSTRUCT{:NonCompliantCompany rdf:type owl:class.
?x rdf:type :NonCompliantCompany;
rdfs:comment "This oil company or service provider
is not compliant because it did not submit the local
content plan within the required window of submission."}]
WHERE{?x :plays-upstream-company-role ?r.
NOT EXISTS{?x rdf:type :NonCompliantCompany}
?x :has-submitted ?z. ?z :at-submission-time ?d.
?w rdf:type :SubmissionWindowLCP.
?w ti:hasIntervalStartDate ?wssd.
?w ti:hasIntervalEndDate ?wsed.
FILTER((?d<?wssd)||(?d>?wsed))}"""]
```

The rule in (7) classifies a company as `NonCompliantCompany` if the date on which that company submitted its LCP comes before or after the window of submission.

Concerning the two companies in our use case, i.e., ENI and Tullow, respectively denoted by the RDF individuals `egepl` and `tg1`, they both comply with rules (6) and (7), i.e., they are *not* flagged as `NonCompliantCompany` after the execution of the said two rules. In other words, they both submitted their LCP, and their submissions were within the stipulated window of submission. Hence, flowing from Section 2.1, they are not liable to pay the penalty of five percent of the value of their proceeds obtained from their petroleum activity. However, the reader is invited to double-check these two rules by re-executing the code available on the GitHub repository as well as to make further tests by changing the submission dates of any one of the companies to fall outside the window of submission, by removing the triple that encodes the submission of an LCP by any of the companies, etc.

6.2. Calculating and Checking Aggregate Values

Performing aggregate functions is necessary to implement ACC within the area of local content and participation in Ghana's upstream petroleum industry. As explained in Section 4 above, SPARQL vocabulary includes functions capable of calculating aggregate values. This section shows how to encode in SHACL-SPARQL the norms reported in (1) above, some of which require the calculation of aggregate values on the management staff of a given company.

However, in order to implement the examples in (1), it is first necessary to calculate:

- For how long the company has been operational in Ghana. The required percentages of Ghanaian management staff employed by companies depend on how long the companies have been operational in Ghana.
- The date from which we might consider Ghanaian employees in the calculation of the requisite percentages. As explained at the end of Section 2.1, we want here to consider only Ghanaian management staff who have been employed for *at least six months* from the start date of the window of submission.

Therefore, first of all, we calculate these values for each company in the ABox. This is performed via a SHACL-SPARQL rule having `sh:order` set to 2.

As usual, before showing the implementation of the rule, the (new) classes and properties used therein are listed and explained as follows:

- `date-of-incorporation` specifies the date on which a company was incorporated. This property is used in the `WHERE` clause of the rule.

- `has-incorporation-duration` is a datatype property having `xsd:duration` as object; it specifies the exact duration a company has been incorporated in Ghana. Note that this property is used in the `CONSTRUCT` clause of the rule.
- `employment-allowable-date` is a datatype property having `xsd:dateTime` as its object; it is obtained by subtracting six months from the start date of the window of submission. This property is needed in order to only consider Ghanaian management staff employed on or before the established six months threshold. This is to remedy the lacuna we identified. Additionally, this property is used in the `CONSTRUCT` clause of the rule.

The implementation of the rule is shown in (8).

```
(8) sh:rule[rdf:type sh:SPARQLRule; sh:order 2; sh:prefixes ...;
      sh:construct """CONSTRUCT {?x :has-incorporation-duration ?di.
                                ?x :employment-allowable-date ?mbeb}
      WHERE{?x :plays-upstream-company-role ?r.
      NOT EXISTS{?x rdf:type :NonCompliantCompany}
      ?x :date-of-incorporation ?dt.
      ?w rdf:type :SubmissionWindowLCP.
      ?w ti:hasIntervalStartDate ?wssd.
      BIND(?wssd-?dt AS ?di)
      BIND(?wssd-"P0Y6M"^^xsd:duration AS ?mbeb)}"""]
```

Example (8) goes through the ABox to identify the date of incorporation of the companies; then, it calculates the *duration* of incorporation of each company by subtracting the date of incorporation from the start date of the window of submission. Secondly, it subtracts six months from the start date of the window of submission and associates the resultant date with the same company. The execution of (8) produces the following results:

- In relation to the start date of the window of submission for the year 2023, which is 01/01/2023, ENI's duration of incorporation is calculated as "P2453DT0H0M0.000S", i.e., 6 years, 8 months, and 3 days, while Tullow's is "P6141DT0H0M0.000S", i.e., 16 years, 4 months, and 16 days.
- Both ENI and Tullow are associated with the same six-month threshold date, which is 01/07/2022 considering the start date of the window of submission for 2023.

Once ENI's and Tullow's durations of incorporation and the six-month threshold have been calculated, the next compliance checks may be implemented; these will assess whether or not the two companies have employed the right percentages of Ghanaian management staff considering their duration of incorporation.

To this end, we first need to calculate the overall number of management staff and the total number of *Ghanaian* management staff employed by each company. These numbers are needed to calculate the percentages of Ghanaian management staff required by the norms.

These numbers are calculated by the next three SHACL-SPARQL rules. The first rule, shown in (9), counts the total number of management staff. On the other hand, the second rule, shown in (10), counts the number of Ghanaian management staff who were employed on or before the six-month threshold date. As explained supra, even though a Ghanaian management employee who has been employed after the six-month threshold is technically a management staff of the company, he or she is not eligible to be counted as a Ghanaian management staff with respect to the submission window for the year under review (2023). Rules (9) and (10) use the following properties:

- `has-total-management-staff` is a datatype property having `xsd:integer` as the object; it specifies the total number of management staff employed.
- `has-total-gh-management-staff` is a datatype property having `xsd:integer` as the object; it specifies the number of Ghanaian management staff employed.
- `wdt:P27` is a Wikidata property <https://www.wikidata.org/wiki/Property:P27>, accessed 17 November 2024 that our ontology sets as a sub-property of TVC's property

hasValue. It is used to specify the nationality held by a given employee within a specific time frame.

- with-nationality is a sub-property of TVC's property withValue. It specifies the nationality with respect to the value in time and context given by wdt:P27.

Rule (9) is implemented as follows:

```
(9) sh:rule[rdf:type sh:SPARQLRule; sh:order 3; sh:prefixes ...;
      sh:construct """CONSTRUCT{?x :has-total-management-staff ?manemp}
                    {SELECT ?x (count(distinct ?z) as ?manemp)
                     WHERE{?x :plays-upstream-company-role ?r.
                           ?x :has-incorporation-duration ?di.
                           NOT EXISTS{?x rdf:type :NonCompliantCompany}
                           ?z :has-employment-role ?a.
                           ?a :within-employment-context ?d.
                           ?d :with-company ?x.  ?a :with-role ?b.
                           ?b rdf:type :ManagementStaff}GROUP BY ?x}"""]
```

The execution of (9) produces the following results (cf. Table 1 above):

- Both ENI's and Tullow's total number of management staff employees is 4.

Rule (10) is also implemented as follows:

```
(10) sh:rule[rdf:type sh:SPARQLRule; sh:order 3; sh:prefixes ...;
      sh:construct """CONSTRUCT{?x :has-total-gh-management-staff ?ghmanemp}
                    {SELECT ?x (count(distinct ?z) as ?ghmanemp)
                     WHERE{?x :plays-upstream-company-role ?r.
                           ?x :has-incorporation-duration ?di.
                           NOT EXISTS{?x rdf:type :NonCompliantCompany}
                           ?z :has-employment-role ?a.
                           ?a :within-employment-context ?d.
                           ?d :with-company ?x.  ?a :with-role ?b.
                           ?b rdf:type :ManagementStaff.  ?z wdt:P27 ?f.
                           ?f :with-nationality :ghanaian.
                           ?a :at-employment-role-time ?ter.
                           ?ter ti:hasIntervalStartDate ?tersd.
                           ?x :employment-allowable-date ?mbeb.
                           FILTER(?tersd<?mbeb)}GROUP BY ?x}"""]
```

The execution of (10) produces the following results (cf. Table 2 above):

- ENI has only 1 Ghanaian management staff member, whereas Tullow has no Ghanaian management staff members. Consequently, while ENI's single Ghanaian management staff member was employed on or before the six-month threshold date, in the case of Tullow, even though the company has three Ghanaian management staff members, all of them were employed after the six-month threshold.

Nevertheless, while the RDF triple “:egepl :has-total-gh-management-staff 1” is indeed created for ENI and added to the inferred knowledge graph, no corresponding triple is created for Tullow: since Tullow does not employ any Ghanaian management staff employees, rule (10) does *not* trigger because its WHERE clause is indeed always false.

To deal with these particular cases, we need to add a further rule, having sh:order higher than (10)'s, that assigns 0 to the has-total-gh-management-staff of a company if the RDF property is not (yet) defined for that company.

```
(11) sh:rule[rdf:type sh:SPARQLRule; sh:order 4; sh:prefixes ...;
sh:construct """CONSTRUCT{?x :has-total-gh-management-staff 0}
WHERE{?x :plays-upstream-company-role ?r.
NOT EXISTS{?x rdf:type :NonCompliantCompany}
NOT EXISTS{?x :has-total-gh-management-staff ?ghmanemp}}"""]
```

Finally, after having calculated the total number of management staff and the number of Ghanaian management staff, it is possible to introduce rules that check whether the Ghanaian employment percentages required by the law are met; companies that do not comply with the requirement are again flagged as `NonCompliantCompany`.

Rules (12) and (13), respectively, check the percentages for companies that have been incorporated in Ghana for more than 5 years but less than 10 years, e.g., ENI, and for companies that have been incorporated in Ghana for more than 10 years, e.g., Tullow. Specifically, in line with what is required by the norms, (12) checks whether the percentage of Ghanaian management staff is less than 50%, while (13) checks whether the percentage of Ghanaian management staff is less than 70%.

```
(12) sh:rule[rdf:type sh:SPARQLRule; sh:order 5; sh:prefixes ...;
sh:construct """CONSTRUCT{?x rdf:type :NonCompliantCompany}
WHERE{?x :plays-upstream-company-role ?r.
?x :has-incorporation-duration ?di.
NOT EXISTS{?x rdf:type :NonCompliantCompany}
FILTER((?di>="P5Y"^^xsd:duration)&&( ?di<"P10Y"^^xsd:duration))
?x :has-total-management-staff ?manemp.
?x :has-total-gh-management-staff ?ghmanemp.
FILTER(?ghmanemp<(?manemp*0.5))}}"""]
```

```
(13) sh:rule[rdf:type sh:SPARQLRule; sh:order 5; sh:prefixes ...;
sh:construct """CONSTRUCT{?x rdf:type :NonCompliantCompany}
WHERE{?x :plays-upstream-company-role ?r.
?x :has-incorporation-duration ?di.
NOT EXISTS{?x rdf:type :NonCompliantCompany}
FILTER(?di >= "P10Y"^^xsd:duration)
?x :has-total-management-staff ?manemp.
?x :has-total-gh-management-staff ?ghmanemp.
FILTER(?ghmanemp<(?manemp*0.7))}}"""]
```

As is easy to understand, the execution of (12) on the chosen use case produces the following results:

- ENI is flagged as `NonCompliantCompany` because it has less than 50% Ghanaian management staff,

while the execution of (13) produces the following results:

- Tullow is flagged as `NonCompliantCompany` because it has less than 70% Ghanaian management staff.

Here, since both Tullow and ENI were both flagged as non-compliant because they did not employ the required percentages of Ghanaians as management staff, they are both liable to pay to the Petroleum Commission the administrative penalty of two hundred thousand penalty units mentioned in Section 2.1.

Once again, we invite the reader to download the code and the use case from the GitHub repository associated with this paper and reproduce the results locally to verify the inferences presented above. We also invite readers to explore the demo by adding, removing, or modifying some of the triples and then re-executing the rules.

7. Conclusions

This paper presented a case study on the regulations governing local content and participation in Ghana's upstream petroleum industry (LI 2204). In his PhD research, the first author analyzes numerous provisions in LI 2204 and other related legislation, which require industry participants to utilize indigenous Ghanaian services, such as legal and financial services. These regulations were then translated into SHACL-SPARQL rules. However, to maintain brevity and focus, this paper does not present all of them, but instead highlights selected rules that showcase two key features of using SHACL-SPARQL for legal reasoning: aggregation and temporal reasoning.

The high expressiveness of SPARQL allows for the modeling of complex regulations requiring aggregate functions and temporal management, while SHACL enables defeasible reasoning through its priority operators.

OWL, on the other hand, is primarily designed for reasoning about relationships and hierarchies, rather than for quantitative operations like counting or calculating. Moreover, OWL lacks native constructs for temporal reasoning and defeasibility, both of which are crucial in legal reasoning. Alternative approaches have therefore been proposed to extend RDF, OWL, SWRL, and other languages with new predicate symbols that introduce specific semantics, as shown in [69,70].

However, we believe that SHACL-SPARQL rules, which are gaining traction within the Semantic Web community [9,64,65,71,72], provide an integrated language with the necessary expressiveness. Since SHACL-SPARQL integrates SPARQL—widely used as the primary formal language for working with RDF data—it could more easily lead to standardized solutions. These solutions may also involve domain experts not necessarily familiar with formal constraints. In other words, while approaches based on OWL and other formal systems have been effective in their respective domains, SHACL-SPARQL's usability and integration of SPARQL provide a unified framework for addressing various aspects that have historically been studied independently.

As the present paper focuses on the *expressivity* needed to represent the norms of our use case, which might be provided by SHACL-SPARQL but not by OWL or other contemporary languages for legal reasoning, it might be argued that, on the other hand, the proposed solution may not be efficient and scalable. Indeed, scalability and efficiency have always been core motivations in OWL research, where many proposed solutions include mathematical proofs to show that computations remain within decidable boundaries to control computational complexity (see [54]). However, as discussed towards the end of Section 4 above, recent publications involving the second author of this paper, in particular [2,66], demonstrate quite the opposite: SHACL-SPARQL rules are significantly more efficient and scalable than most mainstream automated reasoners for the legal domain. Specifically, the experiments in [2,66] show that the ASP-based reasoner DLV2 [67], which can interface with and process SPARQL rules, achieves the best performance, even outperforming other ASP-based reasoners. This paper represents the first attempt to apply the frameworks explored in [2,66], which have only been evaluated using synthetic datasets, to a real-world use case: specific regulations within Ghana's upstream petroleum industry.

Despite the positive outcomes of the research presented here, significant challenges remain in developing an application capable of processing *all* data submitted to the Ghana Petroleum Commission, including the Local Content Plan, the Local Content Performance Report, and other required compliance filing documents. One such challenge arises from the fact that these documents require companies to include *textual* content explaining how they achieved compliance with certain legal requirements outlined in the regulations. The relevant data must then be automatically extracted from this content and formalized in RDF format, prior to processing via SHACL-SPARQL.

To achieve this, we plan to integrate Large Language Models (LLMs) into the final application. We believe that combining LLMs with knowledge graphs is a highly promising yet challenging direction in AI research, and we aim to contribute to this overall effort in our future work. Specifically, following the approach recently adopted in [19], which used

GPT-3 to classify GDPR provisions while fine-tuning the LLM on the corpus from [24], we are now conducting similar experiments by fine-tuning LLMs on the manually crafted SHACL-SPARQL rules developed in the first author's PhD dissertation, including those presented above, within the context of the Innovate UK project Odyssey.

Author Contributions: J.A., as the lead author of the paper, was primarily responsible for drafting the initial version of the manuscript. In addition to his writing contributions, he also developed the accompanying computational artifact, which served as a key component of the research presented in the paper. L.R. and A.Z.W. played a significant role in refining the manuscript. They thoroughly reviewed, edited, and fine-tuned the content to enhance its clarity and scholarly value. Furthermore, both L.R. and A.Z.W. independently tested the computational artifact to verify its accuracy and ensure that it met the necessary technical and research specifications. Their combined efforts contributed significantly to the quality and precision of the final paper. All authors have read and agreed to the published version of the manuscript.

Funding: Funding: This research was supported by the projects “Cost Action CA19134: Distributed Knowledge Graphs” and “Innovate UK project 10106412: Odyssey - Opening the National Archive's legal data to AI for A2J”.

Data Availability Statement: The computational artifact demonstrating the concepts discussed in this paper is available at <https://github.com/kuuku-anim/acc-shacl-li2204>, accessed on 17 November 2024.

Conflicts of Interest: The authors declare no conflicts of interest.

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